SDC Hackathon

Technical Tools Declaration

Hosted by Xiamen University Malaysia (XMUM) AI Club

Submission Deadline: 25 October 2026

Instructions: Declare every API, cloud service, AI tool, dataset, model, and sponsor resource your team used to build the project. Complete every section; where a category does not apply, tick “No” or write “None” – do not leave it blank. Submit as a single PDF document named “SDC2026_TeamName_TechDeclaration.pdf” together with your project submission. Undeclared third-party resources discovered at judging may affect your score or eligibility.

Instagram: @sdc_xmumhack ***Email:*** sdc.xmumhack@gmail.com

1. TEAM AND PROJECT OVERVIEW

Team Name

ctrl + ai

Project / Solution Name

SkillHalfLife - with AI Trend Signal evidence layer

Name (Leader)

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Email & Phone (Leader)

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University Affiliation(s)

University Teknologi Malaysia (UTM)

Live Prototype URL

https://skill-half-life-workforce-dashboard.replit.app/my-cv

Code Repository URL (public / access granted to judges)

https://github.com/amanbulu/SDC-Hackathon-ctrl-ai-.git

2. ARCHITECTURE & TECHNOLOGY STACK

2.1. System Architecture Overview

Describe how the components declared in this document connect end-to-end (e.g., user interface → backend → AI model/API → database → cloud hosting). Attach a system block diagram as an appendix if available.

[Maximum 200 words]

SkillHalfLife's React/TypeScript interface provides synthetic PDF/DOCX intake or a manual questionnaire. Users review extracted skill mentions and confirm their profile before role matching. Browser calls go to the Express API through the application's routed paths.

The backend validates inputs, parses documents in temporary bounded workers and runs a deterministic skills/interest/education matching engine. It combines disclosed SDG preferences with bounded scenario adjustments from eligible stored job signals. There is no embedded or remotely called GenAI model.

Separate adapters retrieve Remotive job postings, arXiv research metadata and OpenDOSM labour statistics. PostgreSQL, accessed through Drizzle ORM, stores normalized public evidence, dates, snapshots and refresh history. Provider cooldowns and source-error states prevent unsupported freshness claims.

The API returns ranked roles, skill gaps and prerequisite-ordered learning steps with references, estimated effort and SDG explanations. Reviewed profiles are tab-scoped; learning progress is browser-local. CV files are not persisted. Replit hosts the web and API services; the verified public deployment is Autoscale. Appendix A supplies the block diagram and dependency inventory.

2. ARCHITECTURE & TECHNOLOGY STACK

2.2. Languages, Frameworks & Libraries

List the main programming languages, frameworks, and third-party libraries that your team did not create. Open-source use is permitted provided it is declared and properly licensed/attributed.

Name & Version	Purpose in Project	License
TypeScript 5.9.3 Node.js 24.22.1	Typed application code; server runtime. Node runtime: MIT and bundled notices.	Apache-2.0; Node: MIT + notices
React 19.1.0 Vite 7.3.7 Tailwind CSS 4.3.3	UI rendering, development/build tooling and styling. Supporting UI packages listed with exact versions in Appendix A.	MIT
Express 5.3.0 Zod 3.25.76	HTTP API, validation and bounded request handling. Express-rate-limit and middleware: Appendix A.	MIT
Drizzle ORM 0.45.4 pg 8.23.1	Relational evidence storage and queries. PostgreSQL server version not verified.	Apache-2.0; pg: MIT
pdf-parse 2.4.5 Mammoth 1.13.0 Multer 2.5.0	Synthetic PDF/DOCX parsing and memory uploads. Yauzl / XML parsing and other dependencies: Appendix A.	Apache-2.0; BSD-2-Clause; MIT
TanStack Query 5.104.1 Recharts 2.15.4 Orval 8.41.0	Typed API data access, charts and generated API clients. Complete direct dependency inventory: Appendix A.	MIT

3. APIs & CLOUD SERVICES

3.1. External APIs

List every external API your solution calls (GenAI, maps, payments, messaging, data feeds, etc.). Do not include API keys or secrets in this document.

API and Provider	Purpose in Project	Access Tier (free / paid / sponsor-provided)
OpenDOSM / data.gov.my	Monthly Malaysian labour-force series [D1].	Free public API
Remotive	Remote-job postings, tags and descriptions [D2]. Endpoint: remotive.com/api/remote-jobs. Attribution, 24-hour feed delay; six-hour application cooldown.	Free public API; provider terms
arXiv API	cs.AI / cs.LG / cs.CL metadata and abstracts [D3]. export.arxiv.org/api/query	Free public API; provider terms
Google Fonts	Browser stylesheet/font delivery for Inter. fonts.googleapis.com / fonts.gstatic.com	Free hosted fonts; Inter: SIL OFL 1.1
GenAI / payment / maps / messaging	None called by the application.	Not applicable
O*NET / UN / learning providers	Linked reference pages, not application API calls. See dataset/content register, Appendix B.	Public web access; provider-specific terms
First-party API	Express /api/skillhalflife/* and /api/trend/*. Internal application endpoints, not external APIs.	Hosted with project

3. APIs & CLOUD SERVICES

3.2. Cloud Platforms & Deployment

List every cloud service used for hosting, storage, databases, authentication, or compute, and indicate how the prototype is deployed.

Service & Provider	Purpose in Project	Plan / Region
Replit	Web / Node API hosting; current published deployment type: Autoscale.	Plan not supplied; region unverified
Project PostgreSQL	Persistent public records, source status, snapshots and refresh history through Drizzle. Workspace-provisioned database; underlying infrastructure provider unverified.	Plan / region / server version not verified
Google Fonts	Hosted stylesheet/font delivery; no application profile is intentionally sent.	Public font delivery; region not controlled
Browser storage	sessionStorage: reviewed synthetic profile. localStorage: learning progress. Not cloud synchronization.	Browser-local; no cloud plan
Authentication service	None integrated for this prototype.	Not applicable
Object/file storage	None used for CV persistence. Uploads are temporarily parsed in memory.	Not applicable
Other cloud / compute	Replit workspace / Agent development assistance. Sponsor resource declared in Section 5.	Replit.ai sponsor resource; plan / region not supplied

4. AI TOOLS, MODELS & DATASETS

4.1. AI Models Used in the Solution

List every AI model embedded in or called by your application, including pre-trained, fine-tuned, and self-trained models.

Model & Version	Source / Provider	Usage (pre-trained / fine-tuned / self-trained)	License / Terms
Rule-based engine (team-authored)	Project source	Deterministic scoring, keyword matching and scenario calculations; not a learned AI model.	Team source; third-party libraries listed separately

4. AI TOOLS, MODELS & DATASETS

4.2. Datasets

List all datasets used for training, fine-tuning, evaluation, or in the running application. State clearly if a dataset contains personal or sensitive data.

Dataset Name	Source / URL	Type (public / private / self-collected / synthetic)	License	Personal Data? (Y/N)
OpenDOSM labour force	OpenDOSM [D1]	Public aggregate running data	CC BY 4.0	N: aggregate
Remotive job postings	Remotive API [D2]	Public postings; not a representative vacancy census	API terms; attribution; not blanket CC	Possible Y: incidental public personal details
arXiv papers metadata	arXiv API [D3]	Public metadata / abstracts; not training data	Metadata CC0; other rights vary	Y: public author names
Occupational references	O*NET [D4]	Public role references; no bulk training import	CC BY 4.0; applicable notices	N intended
SDG definitions	UN SDGs [D5]	Public reference text; app mappings authored by project	UN website terms; not blanket CC	N
Synthetic CVs / test fixtures	Project fixtures	Synthetic inputs and model/parser evaluation	Team-authored	N: fictitious
Skills / roles / scenario catalog	Project model	Team-authored illustrative assumptions; not learned forecasts	Team-authored; references separate	N
Learning-resource link registry	Providers [D6]	Public URL metadata; no copied courses	Provider-specific; links only	N intended

4. AI TOOLS, MODELS & DATASETS

4.3. AI-Assisted Development Tools

Tools used to help write code, design, content, or documentation (e.g., chat assistants, code completion tools, image generators). Declaring these tools does not reduce your score; undeclared use may.

Did your team use any generative-AI tool to help create code, design, text, or other project material?

☑ Yes ☐ No

If yes, list them below:

Tool & Version	Used For	Extent of Use (minor / moderate / major)
Replit Agent (service-managed; version not exposed)	Code, UI/design, debugging, tests, source checks and documentation. Includes platform design/testing helpers.	Major
Google AI Studio (version not specified)	Early system architecture: outlined UI, API and database responsibilities and data flows. Drafted initial components, endpoint structures and matching logic for later review and refinement in Replit.	Minor

4. AI TOOLS, MODELS & DATASETS

Briefly describe how your team reviewed, tested, and took responsibility for AI-generated output:

[Maximum 100 words]

AI-assisted changes were checked through source inspection, TypeScript checks, builds, 22 model/parser tests and browser checks of synthetic intake, confirmation, progress, removal and reset. Resource links, source attribution, uncertainty labels and document layout were reviewed. Deterministic scoring keeps generated prose outside authoritative recommendations. These checks are engineering verification, not independent fairness validation or proof of employment impact. The team remains responsible for reviewing the submitted source, licences and declarations; AI-assisted output is not treated as automatically correct.

5. SPONSOR-PROVIDED RESOURCES

Declare any official SDC sponsor-provided GenAI API, cloud credits, tools, mentoring, or data that your team used.

Did your project use any official SDC sponsor-provided resource?

☑ Yes ☐ No

If yes, list them below:

Sponsor	Resource (API / credits / platform / data)	How It Was Used in the Project
Replit.ai	AI-assisted development platform / tools	Core code development, debugging and data integration.
Replit.ai (continued)	Replit Agent / project workspace. Public datasets retain their original provider attribution.	Connected data adapters, refined processing and scoring, resolved integration errors, and supported further code changes and checks.

5. SPONSOR-PROVIDED RESOURCES

Was the sponsor resource essential to the core functionality of the prototype, or could it be replaced? Explain briefly:

[Maximum 100 words]

Replit.ai was essential to our development workflow, helping debug integration errors, connect public datasets, refine data processing and implement further code requirements. Its AI-assisted workspace supported rapid iteration across the interface, backend and matching logic, followed by checks of the resulting changes. It reduced the effort needed to identify faults and apply consistent fixes as requirements evolved. Alternative tools could reproduce parts of this workflow, but replacing it would require additional setup and redevelopment effort. This development dependence is distinct from the application's rule-based runtime, which does not call a GenAI model.

6. SECURITY, PRIVACY & RESPONSIBLE USE

6.1. Data Handling & Privacy

What user or personal data does your application collect, store, or send to third-party services (including AI APIs)? How is it protected, and how long is it kept? State "No user data is collected" if applicable.

[Maximum 150 words]

The demo accepts synthetic profiles only; real CVs and real personal information must not be entered. User-entered skills, interests and education are processed by the first-party API, not an AI API. Uploaded files remain in temporary server memory; parsing workers terminate and the parent buffer is cleared after processing. Reviewed profiles use sessionStorage; progress uses localStorage until removed/reset or browser data is cleared.

Public evidence persists in PostgreSQL without a configured expiry. arXiv records include public author names; job descriptions may incidentally contain public personal details. No sensitive attributes are intentionally collected. API logs omit bodies, query strings and document text. Platform/Google Fonts requests may expose connection metadata; provider log retention was not verified. Upload limits, validation and synthetic-use attestation reduce risk but cannot prove a document is synthetic.

6. SECURITY, PRIVACY & RESPONSIBLE USE

6.2. Known Limitations & Risks

Identify the main technical limitations of your AI components (e.g., hallucination, bias, rate limits, dependency on a single provider) and any mitigation you have implemented.

[Maximum 150 words]

Matching weights, role thresholds, skill half-lives, SDG weights and practice-hour estimates are illustrative and uncalibrated. Outputs are planning support, not hiring decisions, credentials or job-loss predictions. The engine is rule-based; no generative-model hallucination occurs at runtime, but source text and recommendations can still be incomplete or misleading.

Remotive is a delayed global remote feed, not Malaysia's vacancy census; arXiv research activity is not hiring demand; OpenDOSM aggregates do not measure individual skill demand. Sparse evidence produces insufficient-data labels and zero market adjustment where thresholds are unmet. Outages preserve stored records with freshness/error notices.

Server-side cooldowns, timeouts and bounded parsing mitigate abuse and provider failures. Automatic ingestion scheduling, representative outcome validation and a formal fairness/security audit remain incomplete. No demographic input is used for matching, but this does not establish fairness. Public-source and course-content terms still apply.

7. COMPLIANCE & READINESS

Tick all boxes that apply before submitting.

- ☑ All APIs, cloud services, AI tools, models, and datasets used in the project are declared in this document.
- ☑ All third-party code, datasets, and models are used in accordance with their licenses, and attribution is provided where required.
- ☑ No API keys, passwords, or other credentials are exposed in the code repository, prototype, or this document.
- ☑ Any use of generative-AI tools for development has been disclosed and reviewed by the team.
- ☑ Any use of official SDC sponsor resources has been declared in Section 5.
- ☑ The prototype described here is accessible to judges through the links provided in Section 1.
- ☑ All fields in this template have been completed within the word limits stated.

8. DECLARATION

By submitting this document, the team leader confirms that the information provided is complete and accurate to the best of the team's knowledge, and that every API, cloud service, AI tool, dataset, model, and sponsor resource used in the project has been declared.

[Signature]



[Date]

25/10/2026

Appendix A | Architecture & software inventory

Team: ctrl + ai; five members; University Teknologi Malaysia (UTM). Scope: SkillHalfLife frontend, shared API/model libraries and AI Trend Signal evidence services. Exact installed direct-dependency versions are recorded below; declared-but-unused template dependencies are included conservatively. This is a disclosure inventory, not an independent licence audit.

Browser UI / reviewed synthetic profile	React + TypeScript + Vite
First-party API / parser worker	Express + Zod + Multer + PDF/DOCX parsers
Matching / pathway calculation	Team-authored deterministic engine; no GenAI API
Evidence ingestion -> PostgreSQL	Remotive / arXiv / OpenDOSM -> validation -> Drizzle
API response -> user pathway	Roles, gaps, links, scenarios and SDG explanations
Hosting and local state	Replit Autoscale; browser session/local storage

Runtime observed in the workspace: Node.js 24.22.1 (MIT and bundled third-party notices); pnpm 10.26.1 (MIT). PostgreSQL uses the PostgreSQL License; its server version and underlying infrastructure region were not queried. The live route and catalog API returned HTTP 200 during preparation; that is an availability check, not a full production audit.

Direct dependency	Installed version	Declared licence
@hookform/resolvers	3.10.0	MIT
@radix-ui/react-accordion	1.2.21	MIT
@radix-ui/react-alert-dialog	1.1.24	MIT
@radix-ui/react-aspect-ratio	1.1.16	MIT
@radix-ui/react-avatar	1.2.7	MIT
@radix-ui/react-checkbox	1.3.12	MIT
@radix-ui/react-collapsible	1.1.21	MIT
@radix-ui/react-context-menu	2.3.8	MIT
@radix-ui/react-dialog	1.2.0	MIT
@radix-ui/react-dropdown-menu	2.1.25	MIT
@radix-ui/react-hover-card	1.1.24	MIT
@radix-ui/react-label	2.1.16	MIT
@radix-ui/react-menubar	1.1.25	MIT
@radix-ui/react-navigation-menu	1.3.0	MIT
@radix-ui/react-popover	1.2.0	MIT
@radix-ui/react-progress	1.1.17	MIT
@radix-ui/react-radio-group	1.4.8	MIT
@radix-ui/react-scroll-area	1.3.0	MIT
@radix-ui/react-select	2.3.8	MIT
@radix-ui/react-separator	1.1.16	MIT
@radix-ui/react-slider	1.5.0	MIT
@radix-ui/react-slot	1.4.0	MIT

Direct dependency	Installed version	Declared licence
@radix-ui/react-switch	1.3.8	MIT
@radix-ui/react-tabs	1.1.22	MIT
@radix-ui/react-toast	1.2.24	MIT
@radix-ui/react-toggle	1.1.19	MIT
@radix-ui/react-toggle-group	1.1.20	MIT
@radix-ui/react-tooltip	1.3.0	MIT
@replit/connectors-sdk	0.4.3	See upstream terms
@replit/vite-plugin-cartographer	0.5.21	See upstream terms
@replit/vite-plugin-dev-banner	0.1.2	See upstream terms
@replit/vite-plugin-runtime-error-modal	0.0.6	See upstream terms
@tailwindcss/typography	0.5.20	MIT
@tailwindcss/vite	4.3.3	MIT
@tanstack/react-query	5.104.1	MIT
@tanstack/react-table	9.3.1	MIT
@types/cookie-parser	1.4.10	MIT
@types/cors	2.8.19	MIT
@types/express	5.0.6	MIT
@types/multer	2.3.0	MIT
@types/node	25.9.9	MIT
@types/papaparse	5.5.2	MIT
@types/pg	8.23.1	MIT
@types/react	19.3.0	MIT
@types/react-csv	1.1.10	MIT
@types/react-dom	19.3.0	MIT
@types/yauzl	3.4.0	MIT
@vitejs/plugin-react	5.2.0	MIT
class-variance-authority	0.7.1	Apache-2.0
clsx	2.1.1	MIT
cmdk	1.1.1	MIT
cookie-parser	1.4.7	MIT
cors	2.8.6	MIT
d3-scale-chromatic	3.1.0	ISC
date-fns	3.6.0	MIT
drizzle-kit	0.31.11	MIT
drizzle-orm	0.45.4	Apache-2.0
drizzle-zod	0.8.3	Apache-2.0
embla-carousel-react	8.6.0	MIT
esbuild	0.28.2	MIT

Direct dependency	Installed version	Declared licence
esbuild-plugin-pino	2.3.3	MIT
express	5.3.0	MIT
express-rate-limit	8.7.1	MIT
fast-xml-parser	5.11.2	MIT
framer-motion	12.43.0	MIT
input-otp	1.5.0	MIT
lucide-react	0.545.0	ISC
mammoth	1.13.0	BSD-2-Clause
multer	2.5.0	MIT
next-themes	0.4.6	MIT
orval	8.41.0	MIT
papaparse	5.7.0	MIT
pdf-parse	2.4.5	Apache-2.0
pg	8.23.1	MIT
pino	9.14.0	MIT
pino-http	10.5.0	MIT
pino-pretty	13.2.0	MIT
prettier	3.9.10	MIT
react	19.1.0	MIT
react-csv	2.2.2	MIT
react-day-picker	9.14.0	MIT
react-dom	19.1.0	MIT
react-hook-form	7.90.0	MIT
react-icons	5.7.0	MIT
react-resizable-panels	2.1.9	MIT
recharts	2.15.4	MIT
sonner	2.0.8	MIT
tailwind-merge	3.7.0	MIT
tailwindcss	4.3.3	MIT
thread-stream	3.1.0	MIT
tw-animate-css	1.4.0	MIT
typescript	5.9.3	Apache-2.0
vaul	1.1.2	MIT
vite	7.3.7	MIT
wouter	3.13.0	Unlicense
yauzl	3.4.0	MIT
zod	3.25.76	MIT

Workspace-owned packages are original project modules, not third-party models: frontend, API, skill model, generated clients and schema packages. Additional transitive dependencies are resolved in pnpm-lock.yaml; retain their required licences/NOTICE files.

Package metadata alone does not establish compliance of every asset or dependency.

Declaration-document preparation only: Python 3.13.11; PyMuPDF 2.0 (AGPL/commercial dual licensing); ReportLab 5.1.0 (BSD-style); python-docx 1.2.0 (MIT). These authoring tools are not embedded in the career application.

Appendix B | Public evidence & content register

[D1] OpenDOSM Monthly Principal Labour Force Statistics
https://open.dosm.gov.my/data-catalogue/lfs_month
Endpoint/reference: https://api.data.gov.my/opendosm?id=lfs_month
CC BY 4.0. National aggregates, not skill demand or vacancies.

[D2] Remotive remote jobs API
https://remotive.com/remote-jobs/api
Endpoint/reference: https://remotive.com/api/remote-jobs
Provider API terms, retained Remotive links/credit and delayed records. Application cooldown: six hours.

[D3] arXiv API / terms
https://info.arxiv.org/help/api/tou.html
Endpoint/reference: https://export.arxiv.org/api/query
Descriptive metadata: CC0 1.0; abstracts/papers have applicable rights, not assumed CC0. Application cooldown: one hour.

[D4] O*NET occupational references
https://www.onetcenter.org/license_db.html
Endpoint/reference: https://www.onetonline.org/
U.S. Department of Labor / Employment and Training Administration. Related occupations; no validation of app thresholds or Malaysia outcomes.

[D5] UN Sustainable Development Goals
https://sdgs.un.org/goals
Public goal definitions; project-authored role/skill-to-goal mechanisms. Follow UN website/material use terms.

[D6] Learning-resource registry
Provider URLs below
Public links and limited metadata, not republished courses. Free learning access is not a free certificate, exam or runtime guarantee.

Thank you to arXiv for use of its open access interoperability. Public author names and incidental personal references in public job text are not synthetic user profiles; they must not be used to infer sensitive traits or populate real CVs. No training or fine-tuning dataset is used.

Appendix B | Linked learning materials (continued)

Harvard CS50
https://cs50.harvard.edu/x/

Microsoft Learn
https://learn.microsoft.com/en-us/training/paths/microsoft-azure-fundamentals-describe-cloud-concepts/

PortSwigger Web Security Academy
https://portswigger.net/web-security

System Design Primer
https://github.com/donnemartin/system-design-primer

Elements of AI
https://www.elementsofai.com/

Google Technical Writing
https://developers.google.com/tech-writing

SQLBolt
https://sqlbolt.com/

Cisco Networking Basics
https://www.netacad.com/courses/networking-basics

Google Machine Learning Crash Course
https://developers.google.com/machine-learning/crash-course

Apache Airflow
https://airflow.apache.org/docs/apache-airflow/stable/tutorial/index.html

Resource pages were checked for the prototype on 24 October 2026. Course content is linked rather than copied. Provider-specific terms apply; underlying repository/tutorial licences do not automatically license unrelated materials.

Google Fonts: https://fonts.google.com/specimen/Inter ; Inter font licence: SIL Open Font License 1.1. Hosted font requests involve Google Fonts infrastructure; the application does not intentionally send CV/profile content in those requests.

AI-assisted development: Replit Agent, service-managed version not exposed, was used substantially for implementation, design, testing and submission drafting. Platform design/testing helpers form part of that tool, not separately integrated runtime models. Google AI Studio was used to a minor extent for early system architecture and initial code drafting, including component responsibilities, data flows, endpoint structures and matching-logic outlines. Its exact version/model was not specified by the team; this use does not constitute a runtime model integration.

Sponsor-provided resource: Sponsor: Replit.ai. Resource: AI-assisted development platform, Replit Agent and the project workspace. These resources supported implementation, debugging and integration of public datasets into the application. Assistance included connecting data adapters, refining processing and matching logic, resolving frontend/backend issues, applying code changes and checking outputs. The platform supported continued development as requirements evolved. Replit.ai supplied development assistance; the datasets remain attributable to OpenDOSM, Remotive and arXiv. No separate sponsor-provided runtime GenAI API is claimed.